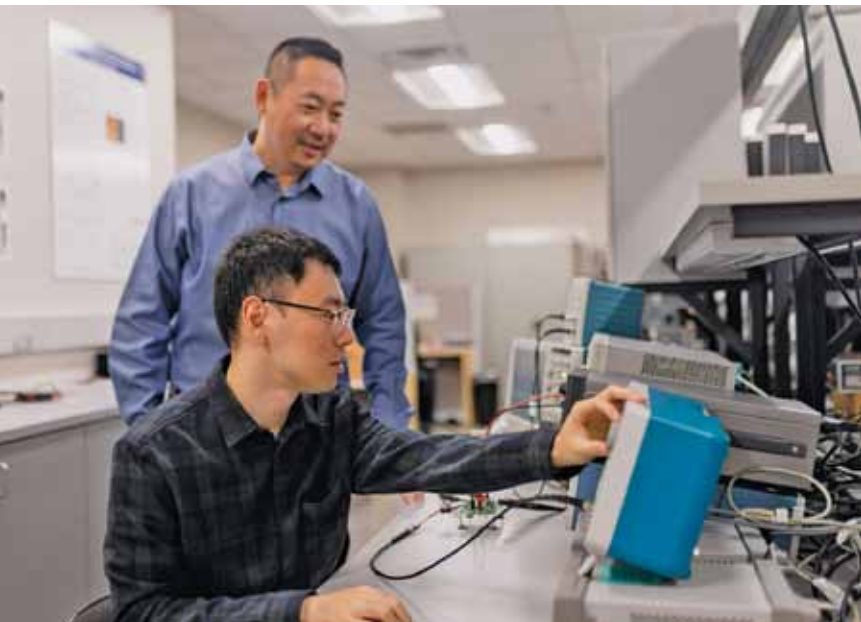


A sensor technology detects electromagnetic interference in EV



Electrical engineering doctoral student Lixiong Du (seated) worked with Dr Dongsheng Brian Ma on chip measurements. The researchers developed technology that uses on-chip countermeasures to control electromagnetic interference (Courtesy: <https://news.utdallas.edu>)

Researchers at the University of Texas at Dallas have introduced a technology designed to identify and mitigate electromagnetic interference (EMI) in electric vehicles (EVs). The team has introduced an intelligent sensor that not only senses EMI but also acts to curtail its effects. This sensor can potentially be embedded in a range of devices, from electric to hybrid vehicles, as well as gadgets like smartphones. This new technology continuously evaluates factors like input voltage and load current, hallmarks of heightened EMI, and swiftly introduces in-chip interventions upon detecting anomalies. The researchers equated this to identifying early signs of conditions like high blood sugar for healthcare. This proactive strategy aims to bolster the safety and efficiency of EVs, ensuring internal EMI doesn't hamper adjacent systems. The team envisions their innovation as a cost-effective avenue to elevate the safety and reliability of EVs.

Solid-state breakthrough for safer EV batteries

Researchers at the University of Maryland have developed a new technology to enhance the safety and energy storage of upcoming EVs and other devices by reducing the risk of battery fires. The research suppresses the growth of lithium dendrites, which are harmful branch-like structures in all-solid-state lithium batteries, obstacles that previously impeded the broad adoption of this promising innovation. A new 'interlayer' design for batteries has been created by the researchers to hinder dendrite formation, potentially leading the way for practical solid-state batteries in electric cars. The team highlights that their approach stands out due to stabilising the battery's interfaces between the solid electrolyte and the anode, and between the electrolyte and the cathode. In this battery design, a fluorine-rich layer is introduced to stabilise the cathode end, and the anode's interlayer is enhanced with magnesium and bismuth, effectively preventing the growth of lithium dendrites.



This new design for a battery 'interlayer,' led by Chunsheng Wang of the University of Maryland, avoids dendrite formation and could allow the production of viable solid-state batteries for electric vehicles (Courtesy: <https://economictimes.indiatimes.com>)